

WEEK 5 PROGRESS UPDATE

Multilingual Speech-to-Intent Detection System

Research Progress Report

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Reporting Period	Week 5	Status	Ongoing — MASSIVE Dataset Deep-Dive Analysis

1. Purpose of This Report

Following the Week 4 update on the Whisper → MuRIL speech-to-intent pipeline, this report steps back from model training results and presents a focused analysis of the MASSIVE multilingual NLU dataset that underlies the intent-classification stage of the project. The goal is to document, in one place, what the 60 intent categories represent, how many examples each intent contains, and how this data is distributed across the five Indian languages used in the current experiments — Hindi, Kannada, Malayalam, Tamil and Telugu.

This report is intended to sit alongside the Week 4 update as reference material for interpreting the per-intent error analysis reported there, since several of the intents that performed poorly ($F1 = 0$) are, as shown below, also the intents with the fewest training examples.

2. Dataset Overview

MASSIVE (Multilingual Amazon SLU resource package for Slot-filling, Intent classification, and Virtual assistant Evaluation) is a multilingual natural-language-understanding corpus released by Amazon Science. It was constructed by localizing the English SLURP corpus (single-shot, spoken-style virtual-assistant requests) into a large set of typologically diverse languages, so that the same underlying set of user requests is available, sentence-for-sentence, in every language. This parallel design is what makes MASSIVE useful for the project's central question: whether one multilingual model can learn a language-independent representation of intent.

Characteristic	MASSIVE 1.0
Languages	51 (52 in MASSIVE 1.1, after Catalan was added)
Domains / scenarios	18
Intents	60
Slot types	55
Utterances per locale	16,521 (11,514 train / 2,033 dev / 2,974 test)
Utterances across all locales	Approx. 843,000 ($51 \times 16,521$); Amazon Science describes the overall release as "over 1 million" utterances.
Indian locales	hi-IN, kn-IN, ml-IN, ta-IN, te-IN (5 of the 51 languages)

Note: *bn-BD* (Bengali) is also present in MASSIVE, but represents Bangladesh rather than India and is therefore excluded from the five India-linked locales used in this project.

3. Record Structure

Every utterance in MASSIVE is stored as a JSON record with the same set of fields across all locales, which is what allows the five Indian-language subsets to be merged into a single training file for MuRIL.

Field	Meaning
id	Identifier inherited from the source SLURP utterance
locale	Language and country code, e.g. te-IN
partition	train, dev, or test split
scenario	Broad domain, e.g. alarm
intent	Specific intent, formatted as scenario_intent, e.g. alarm_set
utt	Raw utterance text
annot_utt	Utterance text with slot spans annotated
worker_id	Obfuscated ID of the localization worker (locale-specific)
slot_method	Whether each slot was translated, localized, or left unchanged
judgments	Independent quality scores: intent, slots, grammar, spelling, language ID

4. Scenarios and the Intent Taxonomy

MASSIVE organizes its 60 intents under 18 broad scenarios. A scenario is the general subject area (e.g. Alarm), while an intent is the specific action the user wants within that scenario (e.g. alarm_set, alarm_remove, alarm_query). This two-level structure is the main source of the “semantically close” confusion pairs discussed in Section 9 and in the Week 4 error analysis.

The 18 scenarios are: Alarm, Audio, Calendar, Cooking, Datetime, Email, General, IoT, Lists, Music, News, Play, QA, Recommendation, Social, Takeaway, Transport and Weather.

Table 1 (Section 7) lists all 60 intents with a short description of what each one means and how many examples it contributes. Rather than repeating that description here, this section moves directly to the Indian-language subset, since that is the primary focus of this report.

5. The Five Indian Locales

The current experiments use five of the 51 MASSIVE locales, all linked to India. These five locales share the same 60-intent taxonomy, the same 18 scenarios and the same 55 slot types as every other MASSIVE language — only the surface text of the utterance changes.

Language	Locale	Family	Script
Hindi	hi-IN	Indo-Aryan	Devanagari
Kannada	kn-IN	Dravidian	Kannada

Language	Locale	Family	Script
Malayalam	ml-IN	Dravidian	Malayalam
Tamil	ta-IN	Dravidian	Tamil
Telugu	te-IN	Dravidian	Telugu

The mix of one Indo-Aryan language and four Dravidian languages, spread across four different scripts, is what makes this subset a useful test of whether a single multilingual encoder (MuRIL) can learn one shared intent space across languages that differ in script, morphology and word order.

6. Per-Language and Combined Dataset Size

Because MASSIVE is a parallel corpus, every locale contains exactly the same number of utterances in exactly the same train / dev / test split. This holds for all five Indian locales individually, and combining them for the multilingual MuRIL classifier simply multiplies each split by five.

Split	Examples per language	× 5 Indian languages
Train	11,514	57,570
Validation / Dev	2,033	10,165
Test	2,974	14,870
Total	16,521	82,605

These combined figures match the dataset sizes reported for MuRIL fine-tuning in the Week 4 update (57,570 train / 10,165 validation / 14,870 test), confirming that the five Indian-language subsets were merged without truncation or resampling.

7. Example Counts per Intent (per Indian Language)

The table below lists every one of the 60 intents together with the number of examples it contributes in each of the five Indian languages, and the combined count once all five languages are pooled. Because the dataset is parallel, the per-intent count is identical across Hindi, Kannada, Malayalam, Tamil and Telugu — the same underlying English utterance was localized once per language, so no language has more or fewer examples of a given intent than another. The “Per language” column therefore applies equally to all five locales, and the last column is simply that figure multiplied by five.

#	Intent	Scenario	Meaning	Per language	×5 languages
1	alarm_query	Alarm	Ask about existing alarms	321	1,605
2	alarm_remove	Alarm	Cancel or delete an alarm	223	1,115
3	alarm_set	Alarm	Create a new alarm	263	1,315
4	audio_volume_down	Audio	Decrease playback volume	255	1,275
5	audio_volume_mute	Audio	Mute audio	337	1,685
6	audio_volume_other	Audio	Other volume-related request	68	340
7	audio_volume_up	Audio	Increase playback volume	362	1,810

#	Intent	Scenario	Meaning	Per language	×5 languages
8	calendar_query	Calendar	Ask about calendar events	233	1,165
9	calendar_remove	Calendar	Delete a calendar event	440	2,200
10	calendar_set	Calendar	Create a calendar event	420	2,100
11	cooking_query	Cooking	Ask a cooking-related question	254	1,270
12	cooking_recipe	Cooking	Ask for a recipe	300	1,500
13	datetime_convert	Datetime	Convert a time/date or time zone	223	1,115
14	datetime_query	Datetime	Ask for the date or time	428	2,140
15	email_addcontact	Email	Add an email contact	68	340
16	email_query	Email	Ask about emails	446	2,230
17	email_querycontact	Email	Ask for a contact's email address	254	1,270
18	email_sendemail	Email	Send an email	447	2,235
19	general_greet	General	Greet the assistant	73	365
20	general_joke	General	Ask for a joke	45	225
21	general_quirky	General	Playful or ambiguous interaction	348	1,740
22	iot_cleaning	IoT	Operate a smart cleaning device	331	1,655
23	iot_coffee	IoT	Operate a smart coffee machine	274	1,370
24	iot_hue_lightchange	IoT	Change smart light colour	244	1,220
25	iot_hue_lightdim	IoT	Dim a smart light	372	1,860
26	iot_hue_lightoff	IoT	Turn a smart light off	273	1,365
27	iot_hue_lighton	IoT	Turn a smart light on	234	1,170
28	iot_hue_lightup	IoT	Increase smart light brightness	48	240
29	iot_wemo_off	IoT	Turn a WeMo device off	74	370
30	iot_wemo_on	IoT	Turn a WeMo device on	66	330
31	lists_createoradd	Lists	Create or add to a list	348	1,740
32	lists_query	Lists	Ask what is on a list	336	1,680
33	lists_remove	Lists	Remove an item from a list	445	2,225
34	music_dislikeness	Music	Express dislike of a song	79	395
35	music_likeness	Music	Express liking of a song	280	1,400
36	music_query	Music	Ask about music currently playing	307	1,535
37	music_settings	Music	Change music playback settings	352	1,760
38	news_query	News	Ask for the latest news	318	1,590
39	play_audiobook	Play	Play an audiobook	75	375
40	play_game	Play	Start a game	311	1,555

#	Intent	Scenario	Meaning	Per language	×5 languages
41	play_music	Play	Play music	453	2,265
42	play_podcasts	Play	Play a podcast	227	1,135
43	play_radio	Play	Play the radio	256	1,280
44	qa_currency	QA	Ask about currency or exchange rates	433	2,165
45	qa_definition	QA	Ask for a definition	232	1,160
46	qa_factoid	QA	Ask a factual question	256	1,280
47	qa_maths	QA	Ask a maths question	235	1,175
48	qa_stock	QA	Ask about stock prices	264	1,320
49	recommendation_events	Recommendation	Ask for event recommendations	321	1,605
50	recommendation_locations	Recommendation	Ask for place recommendations	277	1,385
51	recommendation_movies	Recommendation	Ask for movie recommendations	278	1,390
52	social_post	Social	Post to social media	253	1,265
53	social_query	Social	Ask about social-media activity	262	1,310
54	takeaway_order	Takeaway	Order takeaway food	369	1,845
55	takeaway_query	Takeaway	Ask about a takeaway order	67	335
56	transport_query	Transport	Ask general transport information	316	1,580
57	transport_taxi	Transport	Book a taxi	247	1,235
58	transport_ticket	Transport	Ask about or request a ticket	336	1,680
59	transport_traffic	Transport	Ask about traffic conditions	427	2,135
60	weather_query	Weather	Ask about the weather	437	2,185

Average examples per intent, per language: $16,521 \div 60 \approx 275$. In practice the distribution is uneven: the ten best-supported intents each have 400–450 examples per language, while the ten least-supported intents fall in the 45–80 range (Section 9).

8. Example Counts per Scenario (per Indian Language)

Aggregating the 60 intents up to their 18 parent scenarios gives a coarser view of where the data is concentrated. IoT, QA, Play, Transport and Email are the largest scenarios by example count (each made up of multiple intents), while News, General and Takeaway are the smallest.

Scenario	Intents in scenario	Per language	× 5 languages
IoT	9	1,916	9,580
QA	5	1,420	7,100

Scenario	Intents in scenario	Per language	× 5 languages
Transport	4	1,326	6,630
Play	5	1,322	6,610
Email	4	1,215	6,075
Lists	3	1,129	5,645
Calendar	3	1,093	5,465
Audio	4	1,022	5,110
Music	4	1,018	5,090
Recommendation	3	876	4,380
Alarm	3	807	4,035
Datetime	2	651	3,255
Cooking	2	554	2,770
Social	2	515	2,575
General	3	466	2,330
Weather	1	437	2,185
Takeaway	2	436	2,180
News	1	318	1,590

9. High- and Low-Frequency Intents

The ten intents with the most examples per language, and the ten with the fewest, are listed below. The low-frequency list overlaps heavily with the intents that scored F1 = 0 in the Week 4 2-epoch evaluation (general_joke, iot_hue_lightup, iot_wemo_on, takeaway_query, audio_volume_other, email_addcontact, iot_wemo_off, play_audiobook, music_dislikeness, general_greet), which supports the working hypothesis that class imbalance, rather than language difficulty, was the main driver of those failures before additional training epochs were introduced.

#	Highest-frequency intent	Per language	Lowest-frequency intent	Per language
1	play_music	453	general_joke	45
2	email_sendemail	447	iot_hue_lightup	48
3	email_query	446	iot_wemo_on	66
4	lists_remove	445	takeaway_query	67
5	calendar_remove	440	audio_volume_other	68
6	weather_query	437	email_addcontact	68
7	qa_currency	433	general_greet	73
8	datetime_query	428	iot_wemo_off	74

#	Highest-frequency intent	Per language	Lowest-frequency intent	Per language
9	transport_traffic	427	play_audiobook	75
10	calendar_set	420	music_dislikeliness	79

10. Localization and Data-Quality Signals

Beyond the raw utterance and intent label, each MASSIVE record carries two additional signals that are useful for judging data quality in the Indian-language subsets.

Judgments. Independent annotators score every localized utterance on `intent_score`, `slots_score`, `grammar_score`, `spelling_score` and `language_identification`. These scores make it possible to check not just what intent an utterance carries, but whether the Hindi, Kannada, Malayalam, Tamil or Telugu sentence reads naturally and is actually written in the target language.

Slot method. Each slot value is tagged as translated, localized (replaced with a more natural target-language expression rather than a literal translation) or unchanged. This distinction matters for the Indian languages in particular, since technology vocabulary (alarm, email, volume, WhatsApp, settings) is frequently code-mixed with English rather than translated outright.

11. Key Observations

- MASSIVE assigns 16,521 utterances to each locale (11,514 / 2,033 / 2,974 for train / dev / test), and this figure is identical for all five Indian languages because the dataset is parallel.
- Pooling the five Indian locales for MuRIL fine-tuning yields 82,605 utterances in total (57,570 / 10,165 / 14,870), matching the figures used in the Week 4 training run.
- The 60 intents are not evenly represented: per-language counts range from roughly 45 examples (`general_joke`) to roughly 453 examples (`play_music`), a nearly 10× spread.
- IoT, QA, Play and Transport are the most heavily represented scenarios; News, General and Takeaway are the least represented.
- Because the imbalance is identical across languages, it is a property of the dataset itself and not of any one Indian language — the same intents are hard to learn for the same reason (few examples) in Hindi, Kannada, Malayalam, Tamil and Telugu alike.

12. Limitations

- MASSIVE is a text dataset. It supports the MuRIL intent-classification stage directly, but says nothing about ASR quality, which the Week 4 report identified as a separate source of downstream error.
- The 60-intent taxonomy is fixed and closed-set; real user requests that fall outside it cannot be represented.
- The Indian-language utterances originate from localizing English SLURP requests rather than from independently collected Indian voice-assistant conversations, so idioms specific to natural Indian speech (including code-mixing patterns) may be under-represented.

- The intent-frequency imbalance documented in Section 9 is inherited directly from the source SLURP distribution and could not be corrected without collecting additional data for the low-frequency intents.